

TCC805x MCU BSP

API Specification for Local Interconnect Network

Rev. 1.01 [A]

2021-01-22

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1 INTRODUCTION

This document describes the local interconnect network device driver (hereinafter referred to as LIN driver) of TCC805x MCU BSP. The LIN is a serial communications protocol which efficiently supports the control of mechatronics nodes in distributed automotive applications. For more detailed information of the Local Interconnect Network, refer to ***“ISO standard ISO 17987 Part 1-7”***.

Telechips does not have LIN Bus hardware, but user can operate LIN communication using UART (Universal Asynchronous Receiver/Transmitter) interface hardware and external LIN transceiver. For more detailed information of the UART interface hardware in the TCC805x MCU Sub-System, refer to Chapter 12 Universal Asynchronous Receiver/Transmitter (UART) of the ***“PART 11-MICOM SUB-SYSTEM”*** in the ***“TCC805x Full Specification”***. [1]

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

LIN	Local Interconnect Network
PID	Protected Identifier
UART	Universal Asynchronous Receiver/Transmitter

3 SOURCE FILES

3.1 Location of Source Files

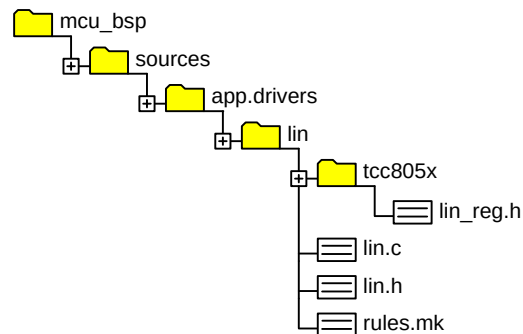


Figure 3.1 Location of Source Files

3.2 Simple Description of SOURCE FILES

- `lin_reg.h`
 - Define variables that are dependent on hardware
- `lin.c`
 - LIN driver source
- `lin.h`
 - LIN driver API header
- `rules.mk`
 - Rules used for build

4 CONSTANTS

4.1 Return Values

This Return values indicate function results.

Declaration

```
#define LIN_OK      (0)
#define LIN_ERROR   (-1)
```

Description

<i>Value</i>	<i>Description</i>
<i>LIN_OK</i>	<i>No error</i>
<i>LIN_ERROR</i>	<i>LIN error occurred</i>

Remark

None

4.2 LIN Data Size

Constant values for maximum size of LIN data

Declaration

```
#define LIN_MAX_DATA_SIZE    (8U)
```

Description

<i>Value</i>	<i>Description</i>
<i>LIN_MAX_DATA_SIZE</i>	<i>Maximum size of LIN data</i>

Remark

None

5 STRUCTURES

5.1 LINConfig_t

This struct includes configuration value for LIN driver.

Declaration

```
typedef struct LINConfig
{
    uint32    uiLinCh;
    uint32    uiPortSel;
    uint32    uiBaud;
    uint32    uiLinSlpPin;
} LINConfig_t;
```

Description

Value	Description
<i>uiLinCh</i>	<i>UART Controller number (0 to 5)</i>
<i>uiPortSel</i>	<i>Channel to Port Mapping Information of UART (0 to 47). This value is saved in Port Selection Register.</i>
<i>uiBaud</i>	<i>Baud rate of LIN communication</i>
<i>uiLinSlpPin</i>	<i>GPIO Port used as LIN Transceiver Sleep Control Pin</i>

Remark

None

6 APIs

6.1 LIN_Init

Initializes the LIN driver, sets the baud rate, UART channel, UART port selection, and wakes up the LIN transceiver.

Declaration

```
sint8 LIN_Init  
(  
    LINConfig_t    sLinCfg  
);
```

Parameters

Value	Description
<i>sLinCfg</i>	<i>[in]</i> The specific configuration. Refer to the LINConfig_t .

Return

Value	Description
<i>LIN_OK</i>	<i>Success</i>
<i>Others</i>	<i>Failure</i>

Remark

None

6.2 LIN_SendData

Sends data to frame identifier.

Declaration

```
sint8 LIN_SendData
(
    LINConfig_t    sLinCfg,
    uint8          ucId,
    uint8 *        pucData,
    uint8          ucSize
);
```

Parameters

Value	Description
<i>sLinCfg</i>	[in] The specific configuration. Refer to the LINConfig_t .
<i>ucId</i>	[in] <i>ucId</i> is a frame identifier
<i>pucData</i>	[in] Data Array for sending. Array size is up to 8.
<i>ucSize</i>	[in] The number of sending data

Return

Value	Description
<i>LIN_ERROR</i>	Failure
<i>LIN_OK</i>	Success

Remark

None

6.3 LIN_ReadData

Reads data from slave that have specific frame identifier.

Declaration

```
sint8 LIN_ReadData  
(  
    LINConfig_t    sLinCfg,  
    uint8          ucId,  
    uint8 *        pucData  
);
```

Parameters

Value	Description
<i>sLinCfg</i>	[in] The specific configuration. Refer to the LINConfig t .
<i>ucId</i>	[in] ucId is a frame identifier
<i>pucData</i>	[out] Data Array for Read. Array size is up to 8.

Return

Value	Description
<i>LIN_ERROR</i>	Failure
<i>Others</i>	The number of read data

Remark

None

6.4 LIN_TransceiverWakeup

Sets LIN Transceiver status.

Declaration

```
sint8 LIN_TransceiverWakeup
(
    LINConfig_t    sLinCfg,
    boolean        bStatus
);
```

Parameters

Value	Description
sLinCfg	[in] The specific configuration. Refer to the LINConfig_t .
bStatus	[in] Set LIN Transceiver operation status. SALEnabled: Set Lin Transceiver as Wake up SALDisabled: Set Lin Transceiver as Sleep mode

Return

Value	Description
LIN_OK	Success
Others	Failure

Remark

None

7 HOW TO USE LIN DRIVER

7.1 Initialize LIN Driver

```
{  
    ...  
  
    LINConfig_t sLinConfig;  
  
    sLinConfig.uiLinCh      = UART_CH2;  
    sLinConfig.uiBaud       = 9600U;  
    sLinConfig.uiPortSel    = 36U;  
    sLinConfig.uiLinSlpPin  = GPIO_GPK(8UL);  
  
    LIN_Init(sLinConfig);  
  
    ...  
}
```

7.2 Write Data

```
{  
    ...  
    uint8 ucSendData[3] = {0xAA, 0xBB, 0xCC};  
    uint8 uiSendId      = 0x11;  
    uint8 ucSendSize    = 3;  
  
    uiStatus = LIN_SendData(sLinConfig, uiSendId, ucSendData, ucSendSize);  
  
    LIN_SendData(sLinConfig, uiSendId, ucSendData, ucSendSize);  
  
    ...  
}
```

7.3 Read Data

```
{  
    ...  
  
    uint8 ucReadData[LIN_MAX_DATA_SIZE] = {0,};  
    uint8 uiRendId      = 0x08;  
    sint8 cReadSize;  
  
    cReadSize = LIN_ReadData(sLinConfig, uiRendId, ucReadData);  
  
    ...  
}
```

8 REFERENCES

- [1] TCC805x Full Specification
- [2] ISO standard ISO 17987 Part 1-7
- [3] Contact Telechips for more details: sales@telechips.com

9 REVISION HISTORY

Rev. 1.01: 2021-01-22

- Updated
 - Chapter 5.1: Declaration
 - Chapter 7.2 and Chapter 7.3: Source Code
- Added
 - Chapter 4.2: LIN Data Size

Rev. 1.00: 2020-11-12

- First version release

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